

TransformED AI

Customer-Centric Improvement Report

Date: 2026-08-26 | Audience: Product & Engineering | Sprint: Pre-Launch Review

This report documents **10 customer-centric gaps** identified through a review of the TransformED AI product specification, sprint reports, CES lifecycle documentation, and decision log as of August 2026. Each finding describes a point at which the current product creates friction, confusion, or a broken feedback loop for the student — and pairs it with a concrete, scoped fix. Findings are ordered by the sequence a new student encounters them, from sign-up through active study. The summary table on page 2 gives an at-a-glance view of effort vs. impact across all ten.

Summary — All Findings

#	Finding	Effort	Impact
1	Generation completion notification	Low	High
2	14-day full-access trial	Low	High
3	Attention monitoring value explanation	Low	High
4	Teach-back feedback response	Medium	High
5	Intervention sensitivity settings	Low	Medium
6	Learner DNA action recommendations	Low	Medium
7	Chapter progress dashboard	Medium	High
8	Failure state UI notices	Low	High
9	Mobile-first pass	High	High
10	Student content quality reporting	Low	High

Key insight: Seven of the ten fixes are Low effort. The biggest single gap is that students never see the loop close — they upload, wait, watch, type, and nothing visible happens in response. Every fix above addresses one point in that invisible loop.

Detailed Findings & Fixes

#1 The Wait Problem — No Notification When Lesson Is Ready

Impact: High

Effort: Low

Problem

Lesson generation takes 5–15 minutes. There is no email, SMS, or push notification when the lesson is ready, and the progress screen does not show named stages — only raw pipeline node names. A student who uploads their chapter and watches a spinner for 15 minutes will not return. The current WebSocket connection requires the tab to stay open, which is an unrealistic expectation on mobile devices.

Fix

Recommended Fix

Send an email (and SMS for Indian users) when the lesson is ready. Replace the raw node-name progress display with named stages: 'Analysing chapter... Building slides... Generating audio...' Allow the student to close the tab and return — the lesson should be waiting for them on the dashboard.

#2 Free Tier Too Restrictive to Demonstrate Value

Impact: High

Effort: Low

Problem

The free tier offers 1 chapter per month with no TTS, no teach-back, and no tutor Q&A.; A new user experiences a silent, text-only lesson with no interaction — the least compelling version of the product. They never see the voice narration, quizzes, or AI interventions that differentiate TransformED from a plain PDF reader. Conversion requires the student to pay before they understand what they are paying for.

Fix

Recommended Fix

Offer a 14-day full-access trial on sign-up before any tier gate activates. The marginal cost of one trial lesson is well under \$3.00 — treat it as acquisition cost, not a giveaway. After the trial, show a clear side-by-side of what the student had vs. what they are losing, with a single upgrade CTA.

#3 Attention Monitoring Value Exchange Is Invisible to Students

Impact: High

Effort: Low

Problem

The consent modal asks for camera access, but students are not told what they get in return. The Composite Engagement Score (CES) is computed silently; the privacy guarantee ('video never leaves your device') is not surfaced on the consent screen. Students who decline consent receive no explanation of what they are missing, and students who grant it see no evidence the feature is helping them.

Fix

Recommended Fix

On the consent screen, lead with the student benefit: 'We use your camera to notice when you are tired or distracted — so we can help before you fall behind.' Display the privacy guarantee prominently on the same screen, not in a linked policy. Show a real-time green/amber/red engagement indicator during the lesson with a one-line tooltip explaining what the colour means.

#4

Teach-Back Has No Visible Payoff for Students

Impact: High

Effort: Medium

Problem

Teach-back is correctly made optional and never gated on score. However, after the student types their explanation and submits, there is no visible feedback response. The scoring is computed and stored, but the student sees nothing meaningful. Teach-back currently functions as a black-box input form, which removes all motivation to engage with it seriously.

Fix**Recommended Fix**

After teach-back submission, show a brief AI-generated response: 'Good — you captured the core idea. One thing to add: [X].' This closes the feedback loop and gives the student an immediate learning signal. The response can be generated by the same mini model used for scoring (gpt-4o-mini) and cached per submission — it does not need to be real-time.

#5

Intervention Messages Feel Like Interruptions, Not Support

Impact: Medium

Effort: Low

Problem

Intervention messages are pre-generated at lesson build time (not dynamic), capped at 3 per session, and triggered entirely by a formula the student never sees. Students cannot adjust their own sensitivity, cannot distinguish between dismissing and engaging with an intervention, and receive no explanation of why one fired. The feature risks feeling intrusive rather than supportive.

Fix**Recommended Fix**

Add an Intervention Style preference in settings with three options: 'Check in with me often', 'Sometimes', 'Only when I am really stuck'. This maps to CES threshold tuning and makes the feature feel like personalisation rather than surveillance. Log which option a student chooses as a signal for the Learner DNA profile.

#6 Learner DNA Is Descriptive But Not Actionable

Impact: Medium

Effort: Low

Problem

The Learner DNA profile correctly avoids clinical scores and uses descriptive language. However, 'descriptive profile only' with no next steps is a curiosity, not a coaching tool. A student who reads their profile learns about themselves but receives no guidance on what to do differently.

Fix**Recommended Fix**

End every Learner DNA profile with 2–3 specific, personalised study suggestions derived from the CES history. Example: 'You tend to lose focus around the 10–12 minute mark — try a 2-minute break at the midpoint of each segment.' This transforms the profile from a report card into a coach.

#7 No Student-Facing Progress or Study History

Impact: High

Effort: Medium

Problem

There is no dashboard view showing which chapters a student has completed, their engagement scores across sessions, or their progress through a book. Without visible momentum, every session feels like starting from zero. Students preparing for exams need to track what they have covered and how well they retained it.

Fix**Recommended Fix**

Add a chapter-progress view per book on the dashboard. Each chapter shows: completion status, CES score from the last session, and date last studied. A simple progress bar across the book ('6 of 12 chapters done') provides the momentum signal. This is one bounded Supabase query per book and requires no new backend logic.

#8 Failure States Are Invisible to Students

Impact: High

Effort: Low

Problem

Several active failure modes have no defined student-facing UI: Imagen 4 Fast is confirmed dead (D121), meaning slides currently have missing images with no notice. The TTS fallback to Browser Speech degrades voice quality silently. If the \$3.00 cost ceiling is hit, lesson quality downgrades with no warning. Silent degradation breaks trust permanently — students assume the product is broken.

Fix**Recommended Fix**

Define an explicit UI notice for each degraded state. Examples: 'Some slide images could not be generated — we are working on it.' / 'Audio quality has been reduced for this session.' / 'This lesson was generated in economy mode.' Brief, honest notices turn a trust-breaking moment into a trust-building one. Each notice should be stored on the lesson record and rendered by the player, not ad-hoc.

#9 Mobile Experience Is Underprioritised

Impact: High

Effort: High

Problem

Sprint 3 patched 2 broken mobile screens reactively. The custom React lesson player and MediaPipe attention tracking were designed for desktop-first. Indian students — the target market — overwhelmingly study on mobile phones. A product that is hard to use on mobile cannot achieve its stated goal of reaching that student cohort.

Fix

Recommended Fix

Schedule a dedicated mobile-first pass before Week 10 launch. At minimum: player controls sized for thumbs; attention monitoring gracefully disabled on mobile with a clear explanation of what the student is missing and why; upload flow tested on mobile Chrome and Safari. Set a mobile breakpoint budget and add a Playwright mobile viewport test to CI so regressions are caught automatically.

#10 No Student Content Quality Feedback Loop

Impact: High

Effort: Low

Problem

Students have no way to flag AI-generated content that is wrong, confusing, or off-topic. A quiz question with an incorrect answer, or a slide that misrepresents the source material, will immediately erode trust — and the engineering team will never know unless students can report it. There is currently no mechanism for this signal to reach the team.

Fix

Recommended Fix

Add a 'Report an issue' button (flag icon) on each slide and each quiz question. Tapping it opens a two-tap flow: select a reason (Wrong information / Confusing / Off-topic / Other) and submit. Reports are logged to an admin queue. Zero backend complexity — one new table row per report. The admin view already exists; add a reports tab.

These findings were derived from the product PRD (v1.0 Final, June 2026), the Decisions Update (June 2026), sprint reports (Sprint 1–3), the CES Lifecycle & Learner Scenarios specification (August 2026), and the intervention copy review. No user interviews were conducted — these are structural gaps in the product design, not usability observations. A follow-up round of student interviews before Week 10 launch is recommended to validate which of these gaps causes the highest real-world friction.